Using Bayesian Linear Regression to Predict Concrete Strength

Group 1 - Jayden Patel, Matthew Martinez,
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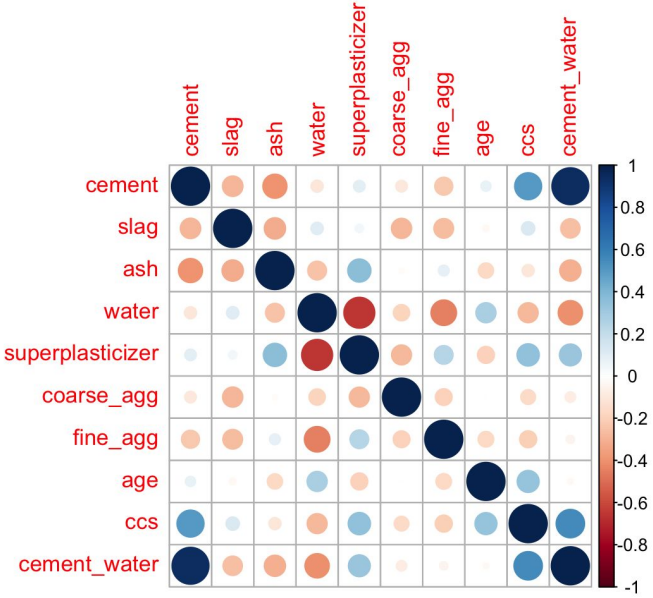
Objectives

- Our goal was to create the most accurate linear regression model as possible to predict concrete compressive strength.
  - 8 variables to choose from
  - 800 observations in the training set
- Our main metric to measure the predictive accuracy was having a low RMSE.
  - Many modifications/transformations made to fit assumptions and increase accuracy
- The following goes through our methodology.

Exploratory Data Analysis

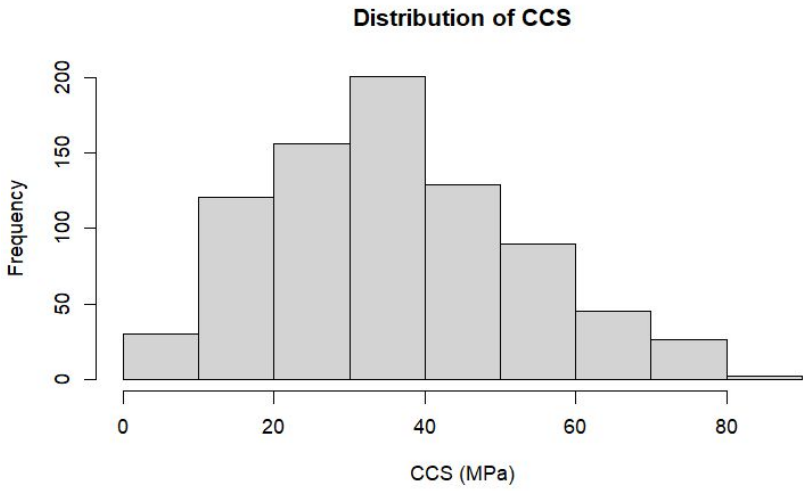


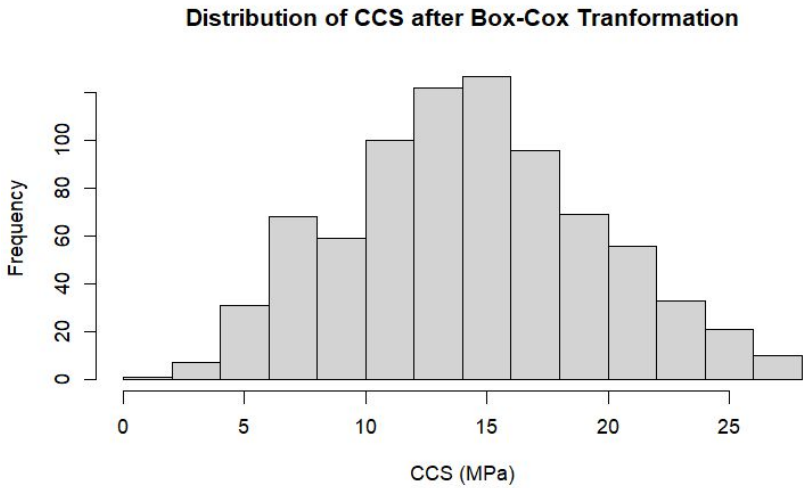
Correlation Matrix on Concrete Dataset

- We used multiple methods to get a feel of the data and its relations to each other before creating the model.
  - Visuals such as scatter plots to compare variables against CCS
  - Histograms to check if CCS fulfills normality
- Through research, we found that the cement-to-water ratio plays a large factor in the strength of concrete.
  - Cement-to-water ratio has the highest positive correlation to CCS (see matrix).

Exploratory Data Analysis

- We found that the distribution of the CCS data was right-skewed.
- A Box-Cox transformation of λ=0.75 made it more symmetric.





Model Specification

```r
concrete_sim <-
  stan_glm(ccs ~ cement_water + ash + slag + age,
           data = new_data,
           family = gaussian,
           chains = 2, iter = 6000, warmup = 1000,
           seed = 303,
           prior = normal(0, 0.5))
```

Final Model

- We specified a Gaussian linear regression model using:
  - cement-to-water ratio
  - ash
  - slag
  - age
- Fitted with the rstanarm package
  - 2 chains for each covariate
  - 6000 iterations, 1000 warm-up
  - No thinning was necessary.
- The other covariates were found to be insignificant and excluded from the final model.
  - A side experiment with an interaction term for cement and water was also found to be insignificant.

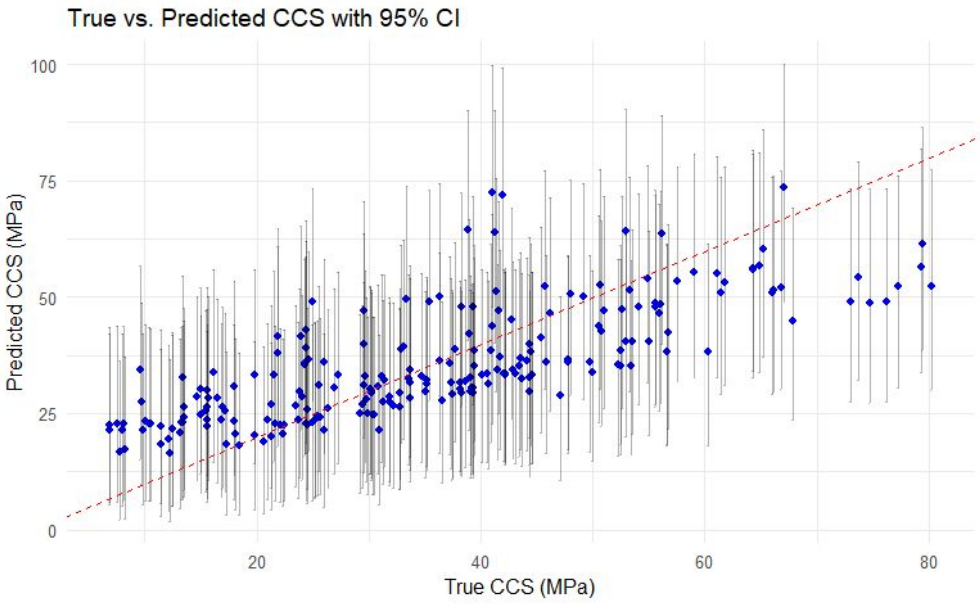
Prior Sensitivity and Model Fit

- To assess prior sensitivity, four distributions were tested with the covariates:
  - Default, autoscaled prior (rstanarm default)
  - Normal(0, 10^2) - very vague
  - Normal(0, 1^2) - moderately vague
  - Normal(0, 0.5^2) - precise
- The precise prior resulted in the best RMSE.
- Convergence was assessed using the Gelman-Rubin test statistic and Geweke diagnostic.
  - All parameters had perfect Gelman-Rubin $\hat{R}$ values.
  - Geweke z-scores were within the ± 1.96 threshold for all parameters.
- ESS for each parameter easily surpassed the recommended threshold of 1,000.

Results

Coefficients:

```
(Intercept) cement_water          ash         slag          age
 1.38547921   5.26805014   0.02419109   0.02433200   0.03374016
```



- All 4 of our chosen predictors had significant 95% credible intervals.
- We found that the cement-to-water ratio parameter had the highest mean (5.27).
- We applied the inverse of the Box-Cox transform to interpret the parameters.
  - For every one unit increase in cement-water ratio, the concrete compressive strength goes up by 9.17 MPa.
- In our testing, our model had an RMSE of 3.347 on a train/test data split.
- Our final RMSE using new test data was 11.073.
- Final interval score was 1.176.

Conclusions

- Cement-to-water ratio is a significant predictor for the strength of concrete.
- Our model was formulated on the significant predictors of CCS after fitting an initial model.
- Using a train/test split on our initial data, the model did well (low RMSE).
- The model did not do as well with the final test data.
    - However, the interval score is fairly low.
- This error may be because the covariates we removed from the initial model were not insignificant for some data points.
- Not likely to have overfitted on training data
    - There were plenty of data points compared to the number of parameters

Questions?

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